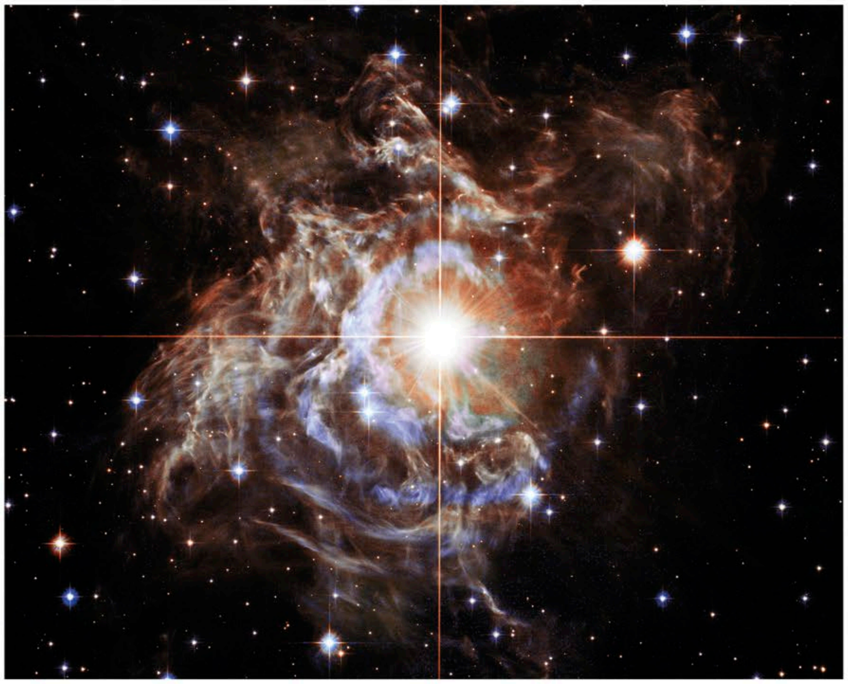
VARIABLE **STARS**

MANY STARS DO NOT SHINE WITH A STEADY LIGHT. SOME OCCASIONALLY DIP OR FLARE IN BRIGHTNESS, WHILE OTHERS SLOWLY PULSATE. THESE ARE EXAMPLES OF WHAT ARE CALLED VARIABLE STARS.

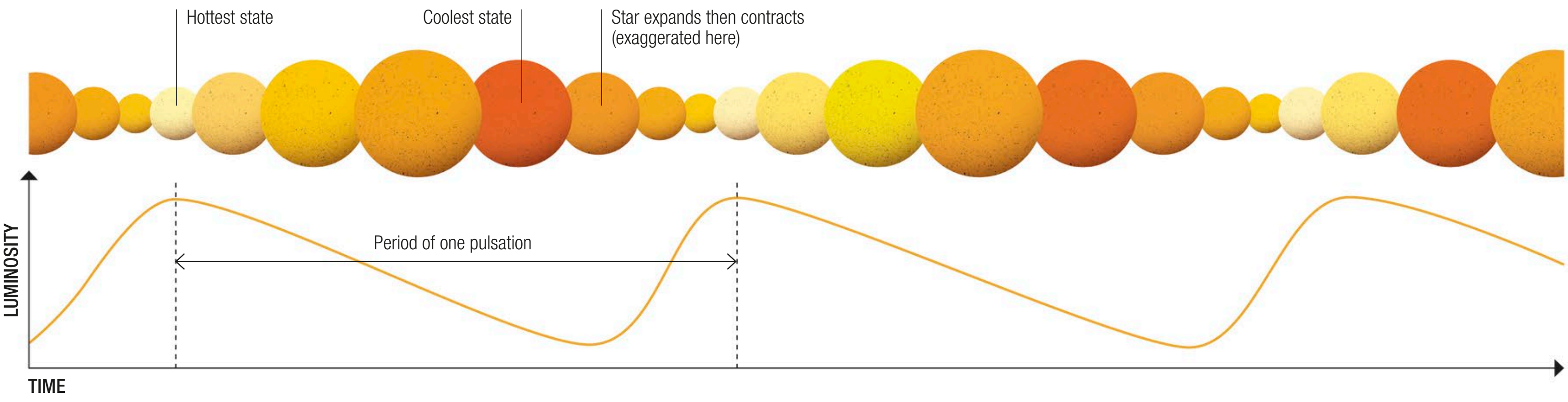
Stars varying in brightness, as seen from Earth, fall into two main categories, called intrinsically variable and extrinsically variable. In intrinsically variable stars, the amount of light emitted by a star varies in a regular cycle, or pulsates, or it occasionally flares up. In extrinsic variables, something affects how much of the star's light reaches Earth.

Pulsating variables

These intrinsically variable stars continuously change in diameter, in a regular cycle, because of fluctuations in the forces that affect their size (see pp.26–27). In a class of stars called Cepheid variables, a close relationship exists between the average light output of the star and the length of its pulsation cycle. This relationship allows astronomers to determine distances within our galaxy and to other galaxies.



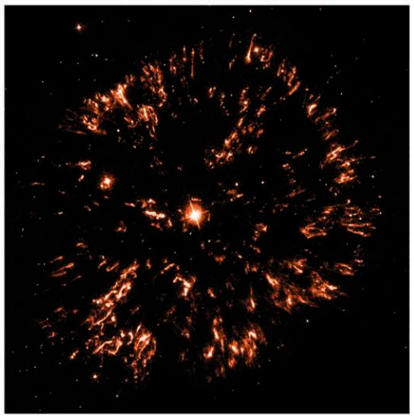
△ **Cepheid variable**
This star, called RS Puppis, a Cepheid variable, varies in brightness by a factor of five in cycles lasting 41.4 days. As this Hubble Space Telescope image shows, the star is shrouded by thick clouds of dust.



△ **Light curve of a pulsating variable**
The amount of light emitted by a pulsating variable fluctuates in a cycle that, depending on the star, can last for anything from several hours to hundreds of days. The fluctuations are closely related to changes in the star's size.

Flaring or cataclysmic variables

Another type of intrinsically variable star, a nova, or cataclysmic variable, is the sudden brightening of a white dwarf star in a binary system (two stars orbiting each other, see p.41). It is caused by a nuclear explosion on the white dwarf's surface. This occurs because the white dwarf's companion star—usually a giant star—has grown so large that its outer layers of hydrogen gas are no longer gravitationally bound to the star and instead fall onto the white dwarf. Subsequently, fusion reactions start up within the accumulated hydrogen on the surface of the white dwarf, triggering a runaway nuclear explosion. Prior to the outburst, the binary system may have been invisible to the naked eye, so the outburst brings the system into visibility as a "nova" (which is Latin for "new") star. Some binary systems produce recurrent novae, separated by quiet periods ranging in length from a few years to thousands of years.



△ **GK Persei nova**
GK Persei has produced a nova about every three years since 1980. Surrounding it is an expanding cloud of gas and dust called the Firework Nebula.

△ **Luminous red nova**
This outburst, from the star V838 Monocerotis, was at first thought to be a regular nova, but it is now suspected be due to two stars colliding.